Corentin Léger - Machine Learning Research Engineer
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Research Engineer with 3 years of experience in Reinforcement Learning, Large Language Models and Multi-Agent systems across academic and industry settings. Particularly interested in RL, LLMs and open-endedness.

Skills

- **Engineering**: Python (PyTorch, SciKit-Learn, Jax), Git, Linux, Distributed Computing (Ray), CI/CD, Hydra
- **Research**: Reinforcement Learning, Large Language Models, Evolutionary Strategies, Multi-agent systems

Experience

- **InstaDeep** — Paris, France
  Research Engineer — *Oct 2025 - Present*
  - Applied Reinforcement Learning research for real-world train dispatching optimization with Deutsche Bahn.
  - Implemented and evaluated single vs multi-agent approaches for different problem formulations (navigation, schedule deviation). Adapted and compared rewards, observations and network architectures across settings.
  - Designed and implemented a parallelized Hyperparameter Optimization pipeline (Ray, Optuna), achieving a 5× speedup over the previous workflow. Enabled reaching 100% navigation metrics on single agent approach for the first time.
  - Improved research infrastructure by optimizing compute usage during experiments, increasing test coverage, and maintaining code quality through regular code reviews.

- **Huawei - Noah's Ark Lab** — Paris, France
  Research Engineer — *Feb 2025 - Oct 2025*
  - AutoML and LLMs research at Paris Noah's Ark Lab, supervised by Balázs Kégl.
  - Developed a tabular data preprocessing pipeline combining LLMs (code generation) and heuristics for feature engineering, as well as feature selection methods to enhance the performance of an internal AutoML tool (Scikit-Learn, Pandas).
  - Co-authored a NeurIPS 2025 workshop paper studying LLM hyperparameter generation skills (LLM-eval), and a preprint on bilevel optimization for Reinforcement Learning.

- **Inria - Flowers Lab** — Bordeaux, France
  Research Engineer — *Dec 2023 - Jan 2025*
  - LLMs and Multi-agent systems research at Flowers Lab, supervised by Clément Moulin Frier.
  - Contributed to two papers (ICLR 2025, preprint) studying text evolution in multi-turn LLM interactions. Built multi-agent simulation, NLP (SpaCy, NLTK) and data visualization tools to run experiments and evaluate metrics (LLM-Culture).
  - Co-developed a framework enabling evolutionary prompt optimization for multi-agent debate (NeurIPS 2025 workshop).
  - Co-developed Vivarium, an interactive multi-agent simulator built in Jax. Achieved real-time communication with Web or Jupyter notebook clients, and used the simulator in a Master's course at UPF Barcelona.

  Research Scientist Intern — *May 2023 - Nov 2023*
  - Meta-Reinforcement Learning research at Flowers Lab, supervised by Clément Moulin Frier
  - Published ER-MRL (EvoAPPS 2024), a method optimizing RNNs with Evolutionary Strategies, in order to improve Deep RL agents' abilities. It enabled to solve partially observable tasks, and adapt faster to unseen environments (Sb3, Gym, Optuna).
  - Co-authored a paper studying the combination of Tabular Reinforcement Learning and Symbolic data (HAL Inria).

- **Connectiv-IT** — Bordeaux, France
  Data Scientist Intern — *May 2022 - Aug 2022*
  - Preprocessed helicopter maintenance data, performing cleaning, outlier detection and imputation (Pandas, Scikit-Learn). Used statistical analysis (SciPy) and clustering to identify key trends in maintenance data.

Selected Projects

- **Open Source Contributions**: Fixed issues in Stable-Baselines3 RL library (10k+ stars). Created a tutorial for parallelized hyperparameter search in ReservoirPy (600+ stars). Contributed to KanRL (interpretability app, PG/PPO implementation).
- **Hackathons**: Won the 2023 and 2024 versions of Hack1Robo (text evolution in populations of LLMs, evolutionary optimization of persuasion capacities). Trained robotic arms with imitation learning and RL at HuggingFace LeRobot hackathon.

Education

- **Ecole Nationale Supérieure de Cognitique** — Bordeaux, France
  Master of Science in Computer and Cognitive Sciences — *Sept. 2020 – Sept. 2023*
  Activities: Bronze medal at French University Volley-Ball Championship 2023

- **Cycle Préparatoire de Bordeaux** — Bordeaux, France
  Bachelor of Science in Mathematics and Physics, Sport-Study contract with ASI Volley-ball — *Sept. 2018 – Jun. 2020*